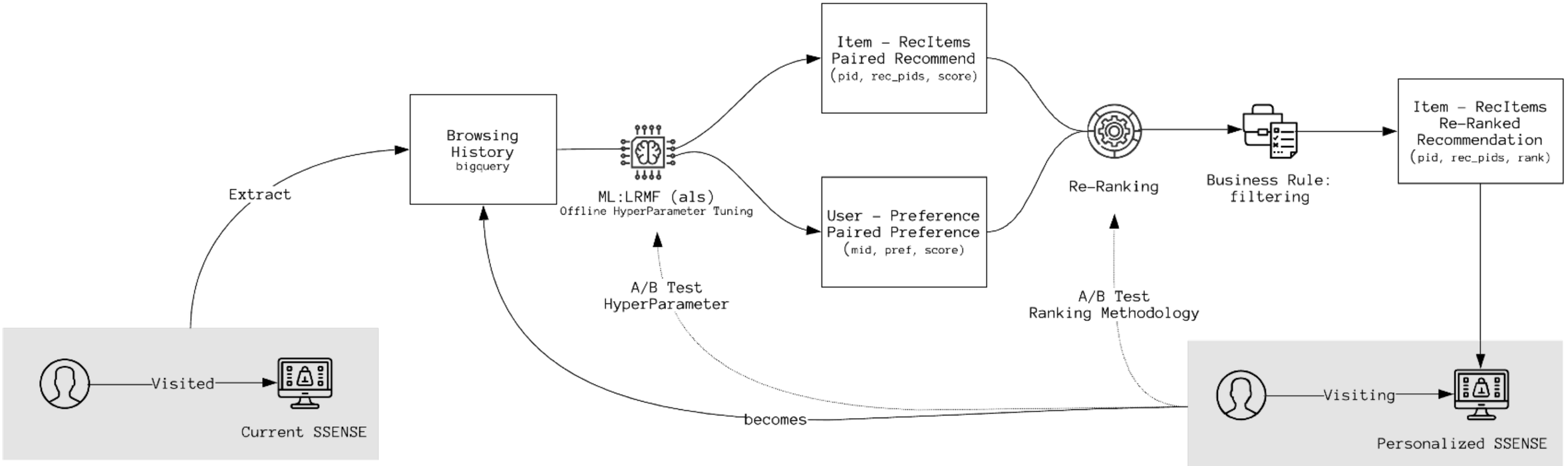
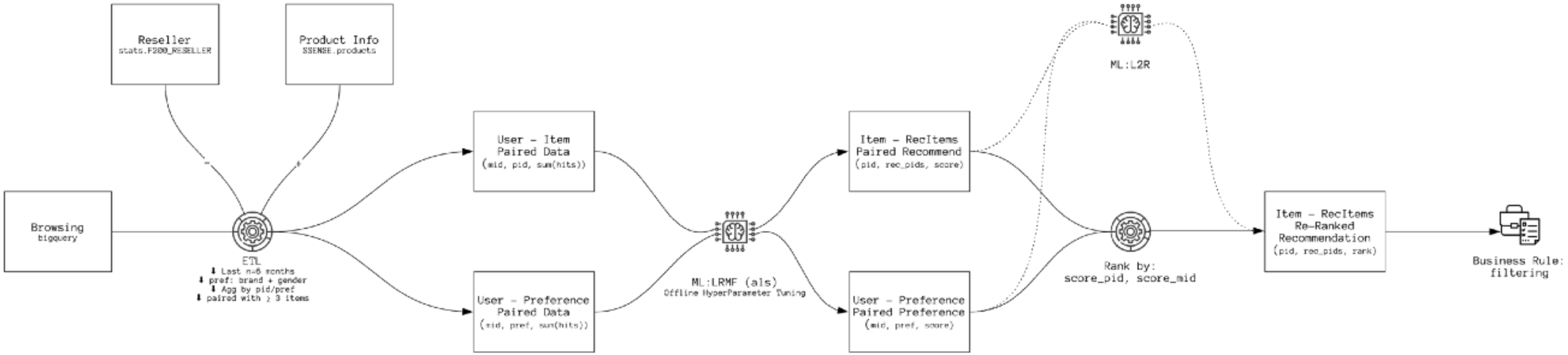
SSENSE

RECOMMENDER SYSTEM

DATA FLOW


Extract
Browsing History
bigquery
ML:LRMF (als)
Offline HyperParameter Tuning
Item - RecItems Paired Recommend
(pid, rec_pids, score)
User - Preference Paired Preference
(mid, pref, score)
Re-Ranking
Business Rule: filtering
Item - RecItems Re-Ranked Recommendation
(pid, rec_pids, rank)
A/B Test HyperParameter
A/B Test Ranking Methodology
Visited
Current SSENSE
becomes
Visiting
Personalized SSENSE

REF: RECOMMENDER SYSTEM ARCHITECTURE



Data Access:

- BigQuery: browsing tuple (mid, pid, sum(hits))
- MySQL Data Science: reseller information (mid)
- MySQL SSENSE: product info (pid, sku, gender, subcategory, active[listed, stock])

Glossary:

ETL: Extract Transform Loading
ML: Machine Learning
LRMF: Low Rank Matrix Factorization
als: alternating least square
L2R: Learning to rank
BPR: Bayesian personalized ranking

SSENSE

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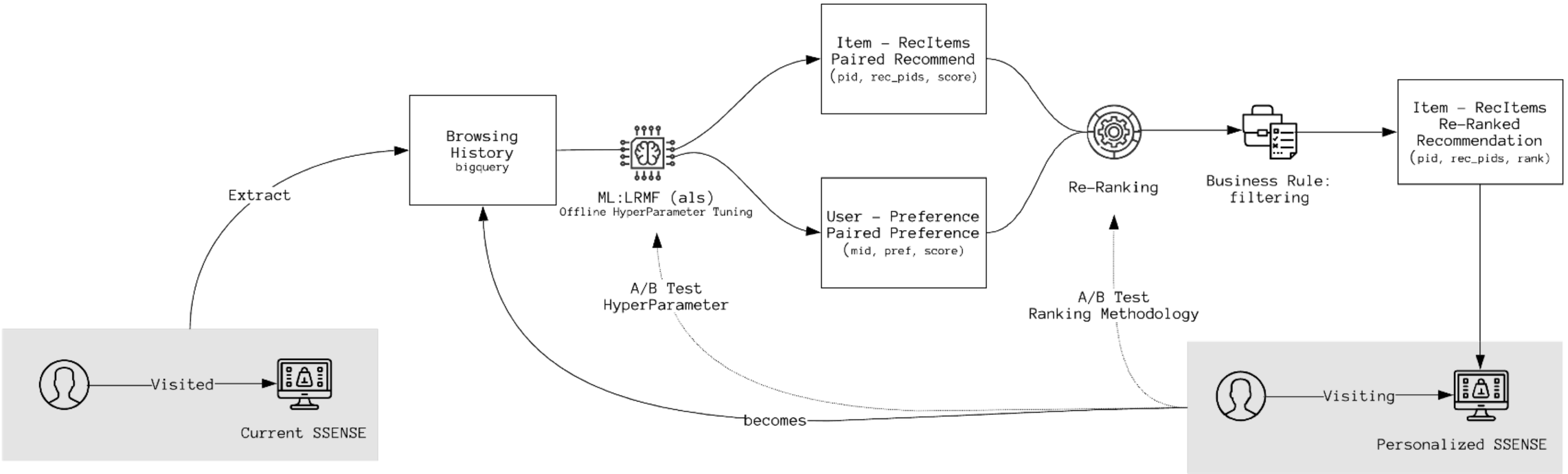
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